

**Causality and entanglement -
Results of the correlation matrix method (CMM)
The sheep hide on the banks of the river**

Abstract

The assumption of Generalized Quantum Theory (GQT) and of the Model of Pragmatic Information (MPI) that Psi is not a transmissible signal, but merely a (measurable) correlation that arises from the entanglement between a person and her or his environment (embodiment) (First Law of the MPI) is also of great relevance in other human sciences. The NT (non-transmission) axiom of the GQT (Second Law of the MPI) makes it understandable why in scientific observations and experimental operationalizations - which include a preparation of the system in which even the NT axiom can potentially be violated - the entanglement correlations existing in the system are weakened to such an extent that the operational criterion for their determination is no longer reached (Third Law of the MPI). At the same time, it is precisely this apparent "lack" that allows the distinction between causality and entanglement in parapsychology and other human sciences. This is achieved by the so-called correlation matrix method (CMM). For this purpose, a matrix (CE) of psychological and physical variables is used to show their correlative relationships and to compare them with a control matrix (CC). The correlations can be the result of causal or entanglement processes.

Using three identical replications of CMM experiments, it can be shown that the data confirm the predictions of the NT axiom in many respects. Thus, it can be seen that in a subsequent direct replication, the entanglement correlations avoid those matrix cells that were occupied in the previous experiment. However, the latter are available for further replication. It also shows that those matrix cells that could potentially contain causal correlations are preferentially available for entanglement correlations, as is to be expected according to the third law of the MPI: "The sheep hide on the banks of the river".

In addition, the CMM is complemented by another method (CP) (physical correlation matrix), which makes it possible to experimentally study the temporal development of entanglement correlations. Since CE and CC contain the same psychological variables, they can be omitted to create a new matrix CP containing only the "physical variables". The experimental matrix CE of the CMM describes an "excitation" of the organizationally closed system (OCS) by the subject. According to this, the OCS is in an "excited (macroscopic) quantum state". However, since the OCS always interacts with its environment (embodiment) - caused by changes in pragmatic information - the system must be subject to a "relaxation process" which is expressed in the control matrix CC; quasi as a "macroscopic decoherence phenomenon". Therefore, CP depends only on the temporal evolution of the system. As in the CMM case, it is possible to distinguish between entanglement correlations

and possible causal correlations (e.g. artefacts). The CMM is illustrated by an overview of all CMM experiments performed so far.

Finally, a single highly significant CMM study has shown that a further prediction of the MPI is fulfilled that refers to the distribution function of the matrix elements. It is to be expected that synchronistic events (in the sense of W. Pauli and C.G. Jung) and entanglement correlations follow a Bose-Einstein-like distribution (because pragmatic information cannot be marked), while causal correlations can be marked and are thus distributed Maxwell-Boltzmann-like.

Key-Words

Generalized quantum theory (GQT), Model of pragmatic information (MPI), Correlation matrix method (CMM), NT-axiom, Cartesian section, Heisenberg section, Exo-Endo discrimination, Organizationally closed system (OCS), Embodiment, Pseudo signal, Macroscopic decoherence, Macroscopic entanglement correlation, Bose-Einstein distribution, Maxwell-Boltzmann distribution

Introduction

I would like to talk about two theoretical models, but above all, about experiments that have been done in recent years and that have to do with "Generalized Quantum Theory" (GQT) and the model of "Pragmatic Information" (MPI). I have deliberately taken "causality and entanglement" as a title because in my opinion it is always seen as opposition, but we will see that it is not opposition at all, but that causality and entanglement need each other to exist at all. There is no entanglement without causality and no causality without entanglement. This is an interesting aspect that can be seen in the experiments.

In the GQT, the basic terms "system", "state" and "observable" are taken from ordinary quantum theory: A system is a part of reality in the broadest sense, at least in principle isolated from the rest of the world and subject to investigation. It is assumed to have the property of being in different states. The concept of state reflects the degree of knowledge an observer has about the system. An observable A of a system is any characteristic that can be studied in (more or less) meaningful ways. The observables A and B can be put together by application of A to B as functions about the set of states. The operation AB performed one after the other is also an observable. The observables A and B are called compatible if $AB = BA$. Observables with $AB \neq BA$ are described as complementary. Complementarity creates entanglement within the system.

The concept of pragmatic information was developed to quantify the meaning of a given piece of information. It is assumed that the (potential) effect that the meaning of the information has on a system can be used for quantification. Von Weizsäcker (1974) suggested that pragmatic information I can be described as the product of two observables, which he called "novelty" E and "confirmation" B.

This approach takes into account that any meaningful information must contain a certain prestructure (confirmation) - for example the native language - in order to be understood by the (receiving) system, but must also contain something new (novelty) in order to effect a change ΔC in the system receiving the information. For example,

an incomprehensible joke in a foreign language would not make anyone laugh (no confirmation), but a joke "from yesterday" would not do so too (no novelty). This also includes the assumption that pragmatic information is not a static but a highly dynamic process, which also manifests itself in the reliability R and the autonomy A of the system. The changes in the system are measured by the changes in the complexity ΔC of the system:

$$I = R * A = B * E ; I = f(C; \Delta C) = n * i$$

The model also assumes that there is a system-specific minimum amount of pragmatic information (or action) i that must be exchanged when an information exchange (measurement) takes place with another system or between two systems. This is simply a different formulation of the unavoidable disturbance associated with a measurement. It is a fundamental assumption of the MPI that a measurement is a preparation of the system.

The assumption of Generalized Quantum Theory (GQT) and the Model of Pragmatic Information (MPI) that causal processes can lead to an entanglement between a person and their environment that goes beyond a purely causal effect can be described by the term "embodiment", a term from cognitive psychology and robotics that is thus also of great relevance in other human sciences (First Law of the MPI). The MPI and the GQT assume that experimental operationalizations always represent a preparation of the investigated system. This means that you have to do something to make an experiment. Thus, it is always an artificial situation and not nature itself. The prepared system behaves in such a way that the NT axiom (also potentially) always plays a role. The NT-axiom (Non Transmission) states that entanglement correlations cannot be used to influence the system in such a way that a predetermined result occurs. This is only possible with causal relationships (Second Law of the MPI).

In GQT and quantum physics, complementarity and entanglement are regarded as one of the main characteristics. But the role of potentiality is also central. It means that potentially existing possibilities that could play a role in an experiment will have an effect even if they are not directly realized. Renninger already called this "negative result measurement" in the 1960s. This means that even if I don't see a photon, but could see it, the interference pattern in the double-slit experiment is still affected. This is a very important insight, because potentiality is something that has not yet been realized and still has an effect. This is the only way to understand why the NT-axiom is so important: Whenever the NT-axiom can potentially be violated, then the entanglement correlations existing in the system are weakened to such an extent that the operational criterion for their determination is no longer reached. They therefore do not disappear without further ado, i.e. entanglement correlations become "ecologically stable". This is the Third Law of the MPI.

At the same time, it is precisely this apparent lack - one always thinks that the NT axiom is something negative, since one cannot do everything one would like - that makes it possible to distinguish between causality and entanglement in parapsychology and other human sciences (cf. e.g. Atmanpacher et.al.2004). It is often the case that a negative statement (think, for example, of the second law of thermodynamics, which states that one cannot build a perpetual motion machine) - even though it is of a very general nature - nevertheless allows very precise

statements. That's to say, one can see at a glance, if something works or not. So we want to ask the question: With what do we actually entangle?

PK-Experiments

I will start with the so-called Schmidt experiments, in which a small random generator with a radioactive source and corresponding electronics converts the results into a binary sequence of numbers and then displays them on a display. A test person sits in front of this display and has the task of wishing that the green lamp on the display lights up more frequently than the red one. The first result was published in 1971 in "New Scientist". It looked as if the subject could influence chance. The subject KG, who believed that she could, got a statistically significant deviation in the desired direction; the subject RR, on the other hand, a theology student who believed that this was not possible, got a significant deviation in the opposite direction. Helmut Schmidt interpreted this result as a psychological condition of psychokinesis (sheep-goat effect). It looks as if a constant force is acting against chance; once in the positive direction - more green lights - and once in the opposite direction. Now the question arises, is this really some kind of force or causal influence, or is it just a correlation? Well, it doesn't look like a correlation. It looks like an effect.

It took me a lot of effort and a long time, I really did the first experiments over thirty years ago (Lucadou 1987, 1991) and did everything to filter out this force or this causal process. The first experiment was based on a very simple idea: I took a strontium 90 source, that's a beta emitter, there's no noise in the usual sense. If the beta particle flies into a Geiger-Müller counter tube, then it's either there or not. So it's not like you can say it could have been there. And then I arranged five Geiger-Müller counters around the source, but connected only one to the display. If PK would be a causal influence and the decay would actually be changed, then all five Geiger-Müller counters have to register this, because the half-life is changed and thus a common influence on the Geiger-Müller counters takes place. But I have noticed that this is not the case. I then used more sophisticated "signal detection methods" for that time to filter out any signals from these results. No signal was detected. But still there was something: There were correlations.

Psycho-physical systems

I define psycho-physical systems as those systems that necessarily require both the physical and psychological language of description to describe them.

Normally this division into the field of physics and of psychology is deliberately chosen, because for example such a term as intention or purpose, for example the intention to make the green lamp light up more often, does not occur in physics. In physics there is no equivalent, you could say there are other categories used. But we all agree that man is a psycho-physical system in such a way that when I rise my arm, for example, there is no doubt that the intention to raise my arm - for whatever reason - is a psychologically describable process. One might wonder why I do this, for example to scratch behind my ear, or to discuss the mind-body-problem, but I always have the problem that I use a descriptive language, namely psychology, to make this relationship clear.

When I now say - and you hear this again and again - that impulses are triggered in the brain that go via my spinal cord into the muscles and lift the arm, then that is of course correct. But of course that's very complicated physics, there's no such thing as intention or will at all. So the physical description language cannot grasp that at all.

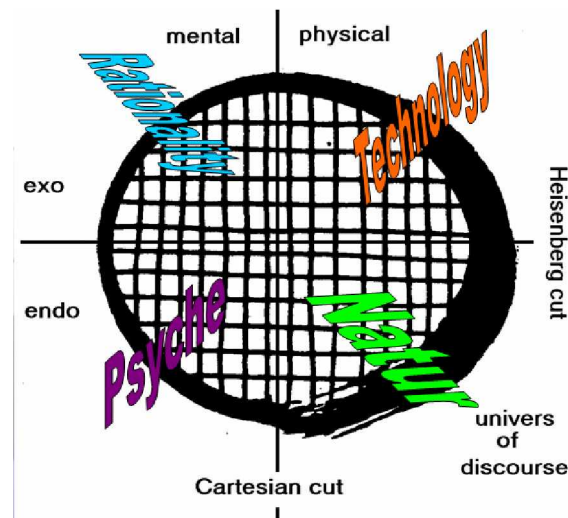


Figure 1: The Cartesian and the Heisenberg-cut

Both description languages exist of course in science and are also used simultaneously, but here it is required that both description are necessary for the systems, I cannot simply omit one. Simple example: When you drive a car, no physicist can tell you where you are going. But if you drive too fast and you fly out of a curve, a physicist can calculate exactly where the car will land.

Figure 1 shows the Cartesian cut - the distinction between the mental and the physical - and the Heisenberg cut, which I will discuss below. The difference between the mental and the physical is indeed a category difference, one that cannot be bridged.

If I assume that the physical directly influences the mental, then the physical must contain a mental quantity. I might be able to imagine that, but I don't know how to put it. Conversely, it's even worse. If I now say that the mental influences the physical, so imagine that I raise my arm, and I actually raise it, then there must be something physical about the thought that triggers the nerve impulses, but that would be something physical again and nothing mental; so there's no to and fro.

Of course many philosophers have already noticed this, and therefore Leibniz's idea of parallelism comes from the fact that one simply cannot reconcile the categories. But we will see: It still works. We know that it has to work. But we don't know how, because we don't have the possibility to describe it in one description language. Now the Heisenberg cut comes into play. The Heisenberg cut is the separation between the epistemic part that is between what I can observe and describe in an experiment. That is something constructed, and that's a very important point. In quantum physics, therefore, we speak of observables. Observables mean it's a quantity I can measure, but I have to do something about it. Now when I say I want to measure the length of this table, I have to do something. I have to take a tape measure, put it on and see how long the table is. Or if I want to determine the weight of the table, I have to put it

on a balance. It has no weight by itself, so to speak. You always have to think about that, because that's one of the basic statements of quantum physics. In order to measure an observable, I always have to do something. That is always an action in reality. And this action projects the observations, the states that describe the observables, into the epistemic world, so to speak. In physics this is quite clear: all observed things are epistemic in this sense, and the question is, what is it like in nature itself?

There is also the distinction between the exo- and endo- description, which essentially go back to Hans Primas (1990, 1992). He assumes that the endo-world really exists, that is the ontic realm. This means that nature also exists without us. We notice that from the fact that once there were dinosaurs. All constructivist ideas say that we produce the world itself. You cannot falsify this, but you get difficulties at this point: Why should there have been dinosaurs when we weren't even there? This is roughly the argument of Primas, and he assumes that whenever I make a measurement in physics or an observation, then I project a measurement from the area below the Heisenberg cut (see figure 1) into the epistemic area. This is not nature itself, but a product of my measurement, so to speak. And you can see this directly in physics, too, when you speak of a "dressed object". This is, so to speak, an "dressed" object. For example, an ammonia molecule, this actually looks different, depending on whether I look at it as an electron configuration or as a nuclear structure. This means that the object I'm looking at is created by a measurement, and then it's in the epistemic region described by a so-called W^* algebra.

Initially we have no direct access to the ontic area. A simple example: If I throw a stone, then I can calculate exactly the trajectory. But of course the projectile itself doesn't calculate, it doesn't have a calculator and calculates its trajectory, it just flies. In other words, nature doesn't need models. Nature simply "exists". And yet this ontic area is not without structure. It is described by a so-called C^* algebra and in this algebra it is actually the case that this area is completely causal and entangled, namely bidirectional. But it is not accessible to us; we cannot look into it without measurement.

The solution to the riddle is that if we now assume that a similar structure exists on the mental side, namely the distinction between the endo-system, the unobserved consciousness, i.e. the unconscious or preconscious, then I also get a Heisenberg cut. The waking consciousness would then correspond to the exo-system.

That is, if we are on the physical level, we can say: Here, in the endo area, we simply call this "nature". And in the mental realm we call it "psyche" or "soul". I'm still there when I've lost my consciousness, when I'm unconscious but not dead, for example. But this is only a description on the side of consciousness. But in the endo area there is actually no Cartesian cut, because nature itself does not make a Cartesian cut. Nature is simply as it is. And of course we and our consciousness are part of nature. So I would say, there is no Cartesian cut here in this area, because the Cartesian cut is, so to speak, the application of categories to the description, and thus an observation.

It is probably the case that animals also possess a kind of consciousness, perhaps even self-confidence, but they are, so to speak, even closer to this endo-area,

because they probably do not have any self-reflection, which would correspond to a "measurement" of the unconscious in the mental area.

But if you talk to many people who have had paranormal experiences or who speak of intuitive experiences, then you get the impression that these are "dives" or "fishing trips" in the endo area that bring unconscious, but also foreign-conscious or even "the ontological" to light. At the moment it is brought to light, it is of course already observed, i.e. we are back in the exo area, and there is no getting around it. Nevertheless, you can say that there is obviously something like a "memory" or "dreamlike structure" that remains of the endo area in the epistemic realm. How does this express itself? In physics these are the remaining entanglement correlations. These can only be prepared with extremely complicated and sensitive experimental arrangements. So you have to do an enormous amount of work to measure entanglement correlations, but that doesn't mean that they are not primarily present, but they don't really "belong" in the exo area. They are something like a kind of "rescued relationship" that is normally lost through observation, because the "superposition" changes into a state of its own through measurement. However, the entanglement correlations rescued in the exo area can be examined more closely, because they have certain properties that we are concerned with here.

If you learned at school that a hydrogen atom consists of a proton in the middle, around which an electron flies in a circle and is held on its orbit by the electrical attraction, then you have to realize that this atom cannot be stable and immediately shatters. We would not have stable matter if we only allowed causal forces. This means that even the simple hydrogen atom is an example of the fact that it cannot exist without quantum physical entanglement. The stability of matter, which for us is so self-evident, by no means self-evident, can only be guaranteed if the causal forces are accompanied by entanglement effects or, as one say, "exchange forces". One can also say that a hydrogen atom is an "organizationally closed system" in which causal and entanglement contexts support each other.

Organisational Closure



Figure 2: Illustration of the "Organizational Closure"

The term "Organizational Closure" (OC) comes from Francesco Varela (1981). He was one of the first biologists to discover before GQT that it is probably better to describe biological processes through the structures of quantum physics. He provided a rather complicated definition of organizational closure, which will be illustrated here in simplified form by Figure 2. Essentially, it is the statement, which naturally applies to physics, that any object that occurs in nature has a surface that it creates itself. One can distinguish between outside and inside, exo and endo. Figure 2 shows completely separated systems. In the next step, they are stimulated to communicate by a given boundary (seminar room). This is of course something dynamic, but it is not yet an entanglement. Entanglement only occurs when the system creates its own surface "from within". Seen from the outside, the organizationally closed system is no longer separable (without destroying it). An example would be a close mother-child relationship, a partnership or a love relationship. This is often such that it is no longer possible to distinguish where the origin of the respective mental states is; they are jointly experienced mental states.

From the outside, the system is an exo-system, but if I belong to the system itself and, for example, am a member of a social group, for example a religion, then there are many interactions within the system that I interpret as causal, but which may represent entanglement. And that is why I introduced the term pseudo-signal.

Pseudo-signals

Pseudo-signals are correlations in the endo-description which appear as deterministic signals, but which in the exo-description represent only non-local correlations (Lucadou 2002). This applies, for example, to many medical applications, the models that are made are causal models, but in reality, of course, an organizationally closed system is formed in the therapeutic relationship or in a therapeutic ritual, and the participants can, from their point of view, regard what happens as causal, but if one looks at it from the outside, one finds that it represents "only" entanglement correlations.

If I now consider this, then I notice that certain characteristics occur here that show that there are no causal connections.

An example would be the swarm behaviour of a swarm of starlings. There is of course a biological reason why the starlings make such beautiful movements as swarms: The aim is to ward off predators. The result is a huge amorphous figure that flies back and forth. When a buzzard comes, he sees the whole figure as an organizationally closed system from the outside and "can't do anything with it" and moves his way, while the individual star is protected in this way, although he doesn't even know - anthropomorphically speaking - what he's fabricating. The flight movements of the individual starlings (local observables, see below) are thus interlocked by the target-oriented behaviour of the swarm, the defence (global observable).

One of the pioneers of collective behaviour was Konrad Zuse, a computer pioneer from Bad Hersfeld. He has published an interesting book on this subject, which is almost forgotten: "Der rechnende Raum" (Zuse 1969). In it he shows that one can model such collective things with cellular automata, and already then he comes to the

conclusion - and he explicitly discusses this - that quantum phenomena can also be described with it.

Local and global Observables

It is crucial that the order of the observables plays a role. If you measure the length and the weight of e.g. a table, then I notice that it doesn't matter whether I measure the length first and then the weight or vice versa. However, this is not the case with many quantities in physics that appear quite harmless, such as location and momentum. In psychology it is almost the rule that observables cannot be interchanged and are therefore called "complementary". Psychologists know that but they believe that this has no consequences. One of the main consequences is that entanglement necessarily occurs. This means that, independent of space and time, pattern coincidences - or, more precisely, correlations of spatially and temporally separated measured values occur.

Typical candidates for complementary observables are local and global observables (see above). However, one must first determine empirically whether they are commutative or not. And only when this has been checked one can legitimately say that complementarity exists and that entanglement correlations can occur.

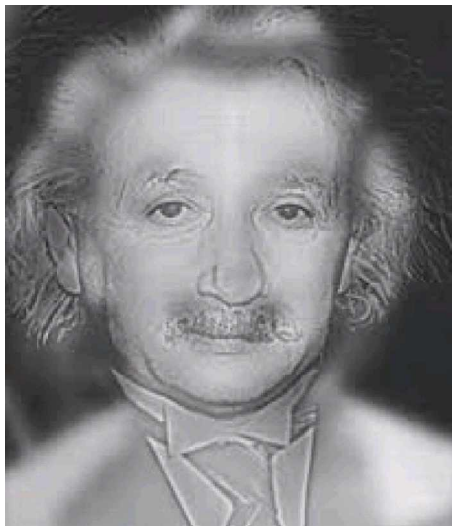


Figure 3: Illustration of a "global and local observer".

In Figure 3 you can observe this directly: If you look closely at the image, you see Albert Einstein (local observable) and if you don't look so closely - for example by blinking your eyes - you see Marilyn Monroe (global observable). The global view is blurred in relation to the local one. Both images, however, are created only in your consciousness. Neither Albert Einstein nor Marilyn Monroe is really depicted. Everything happens only in your mental representation: Only because you know both faces, they are created in your consciousness. So this is really a real phenomenon of consciousness. Mental processes in this sense are global observables, and the neuro-physiological processes are complementary local observables that are entangled. It follows, however, that they cannot be derived from each other. The statement that two complementary observables are incompatible implies that one cannot be derived from the other.

The NT-Axiom

Entanglement correlations cannot be used to transmit signals or causal influences. It is important to note that this also has a potential effect. When I design an experiment so that I could use entanglement correlations for control - even if I don't do it at all - I have already violated the NT axiom. This is the difference to all other models, for example from Dick Biermann, when he says, for example, that you cannot create time paradoxes. So here we go one step further. In other words, time paradoxes are those where I travel through time, kill my own grandfather, and then I create a contradiction because I'm not allowed to exist. So a model must not contain time loops that allow for intervention paradoxes. But it is one step worse, because the NT-axiom says it all: If I travel back in time and could kill my grandfather but don't, I can't exist either.

You can formulate it that way: The decline effect, in parapsychology and elsewhere, is a kind of compromise between the NT axiom and the tendency of organizationally closed systems to remain alive or exist. The organizationally closed systems have, anthropomorphically speaking, a great interest in maintaining their entanglement. Entanglement is elusive if I want to measure it, but it is absolutely necessary and omnipresent, otherwise we would not have stable matter. It is always a compromise. That is why it would be wrong to say that the NT axiom destroys everything, but only destroys the effects to such an extent that there can be no contradiction. If, for example, I now have a sequence of numbers as in the Schmidt experiment, and I establish certain regularity, for example four zeros in succession, and I repeat the experiment, then four ones occur and no longer four zeros. And if I then repeat it again, then there is another pattern, and the pattern is always such that the criterion, which I have previously defined as a signal, is no longer reached. It's not that it disappears completely, but it only disappears to the extent that the NT axiom is not violated.

This can now be generalized to three juggling balls in entangled systems and elsewhere. The NT-Axiom can also be formulated in this way: No matter which system I look at, if I look for a certain entanglement effect in medicine or psychotherapy, it is limited. I can't make it any size unless it's a purely causal effect. But what determines an effect? I call this "documentary quality". The better and more precisely I can describe or document something, the more I can view it as a causal process.

You can really tell: The quality of the documentation is quite bad for those events that are really impressive - such as poltergeist- or haunting phenomena or metal bending. And then you get frustrated afterwards and ask yourself: "Well, I saw it now, but what did I actually see? That's the "uncertainty relation" that the NT-Axiom imposes on us.

This can also be seen in a meta-analysis, which was much disputed. One of the authors (H. Bösch) asked me by e-mail what I could predict based on the MPI. I wrote him back that I expected it to have a decline effect proportional to $1/\sqrt{n}$, where n is the number of trials per experiment. Then he wrote me, he doesn't believe it, he thinks there is a linear decline effect. But this is a direct consequence of the NT-axiom. Figure 4 shows the theoretical prediction and results of all PK experiments made up to this point in 2006 (Bösch et. al. 2006). The red lines are the limit given by the NT axiom. Now, of course, many critics say that this is simply the regression to

the mean. But the slope of the curve is larger. This means that you start with a rather large effect with small studies (n), repeat it and then "slide" down this curve. If you make the n too big, nothing will come out.

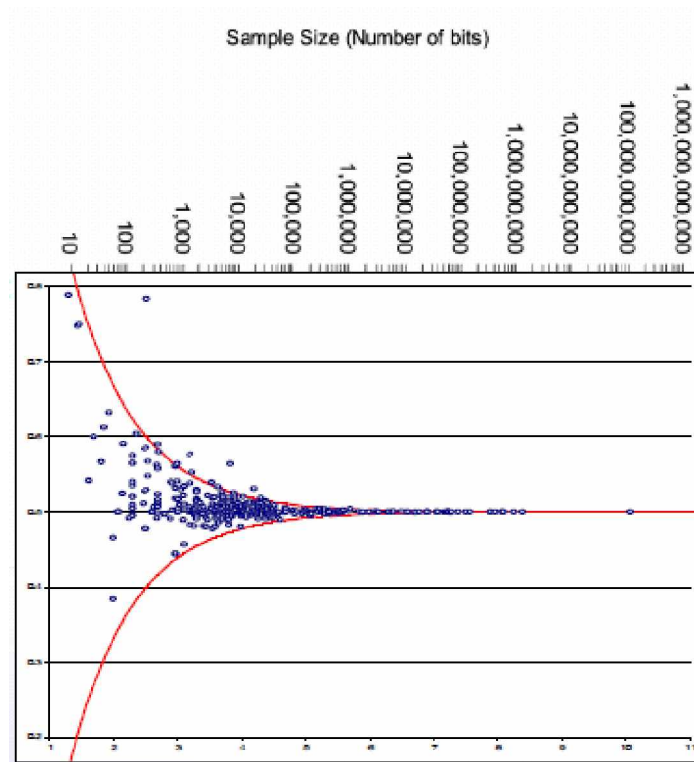


Figure 4: Funnel-plot of PK experiments

And the following was also concealed: There was a meeting, one sat down before the evaluation and asked which result one could expect. I looked at the experimental design and said, "Okay, the experiment is a more or less direct replication of the Princeton studies, with a very large n." I was very impressed. Now we use the NT axiom and it follows that we won't see any effect anymore. But since nature is not malicious - the effect will appear somewhere else, namely in the variance. I predicted that, and then it was checked, and it came out like that. But that was not mentioned in the publication. I find that a strong piece. Whenever a theorist has the possibility to predict something explicitly, this is at least a confirmation for the theoretical model.

If one considers this dependence of the effect strength of $1/\sqrt{n}$ in the meta-analysis, then one can show that it becomes highly significant again. This was shown by Ulrich Timm (2007).

The difference between a signal and an entanglement correlation

The difference between a signal and an entanglement correlation can be clearly seen at Brunswick's lens (1956): It says that if I want to investigate a context in psychology or in the human sciences, it is advisable not to consider just one variable, but to summarize many. If you have observed a person over a longer period of time and got to know him in different situations, you can predict what he will do in a certain situation or how he will fill out a certain questionnaire, rather than looking at just one situation. This is a factor model, as can be seen in Figure 5, for example,

extroversion and extroverted behaviour in individual items. But if you combine many different factors or categories, you get a higher correlation.

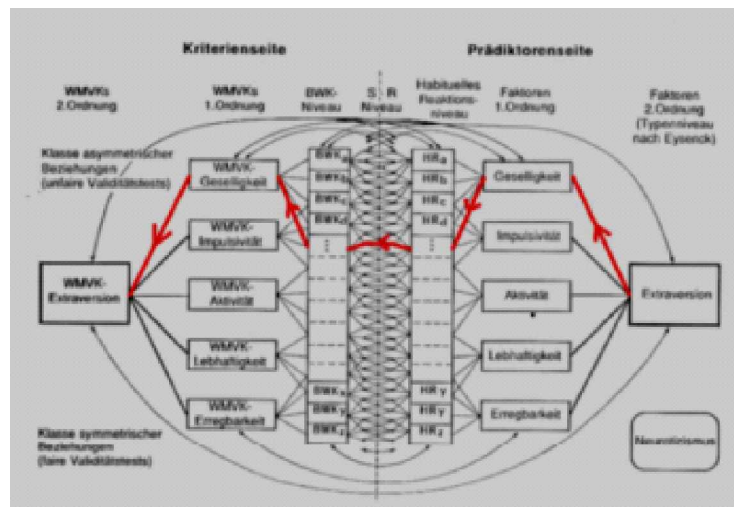


Figure 5: Brunswick's Lens

The idea here is that there are causal relationships that wind their way through this system, so to speak. With entanglement correlations you have a completely different situation. There you have an organizationally closed system and a certain description, e.g. extroverted behaviour, and then the description of someone who is extroverted. The two descriptions correlate, but you cannot specify a path. You can only tell that it is - as long as they are entanglement correlations.

The Correlation Matrix Method (CMM)

This so-called Correlation Matrix Method (CMM) follows exactly this pattern: Here you have a matrix that lists certain physical and psychological observables of the experiment (CE), and there is also a corresponding control matrix (CC) in which no test person was present. Then you see what the difference is between the two. I want to show the whole thing with an example:

Figure 6 shows the correlation matrices of the Viadrina study (Walach et.al. 2014, 2019) with 503 participants. The psychological variables in this case are simply keystrokes. The test subjects were given the task of looking at a beautiful display on the screen where a "random walk" could be observed. There was no talk of "psychokinesis". The instruction was: Try pressing the left and right buttons so that you intuitively steer the process on the screen in the given direction. Then the numbers of keystrokes was recorded and at the same time the different statistical values describing the random process. But the random process and these keystrokes have nothing to do with each other except that every time a key is pressed, the next random process is called. But otherwise they are isolated from each other.

What you see here in this case are 27 psychological variables, I won't go into the details now, and 18 physical variables. If you count the hits according to the classical evaluation, there are no extra hits at all. So there is no deviation from chance.

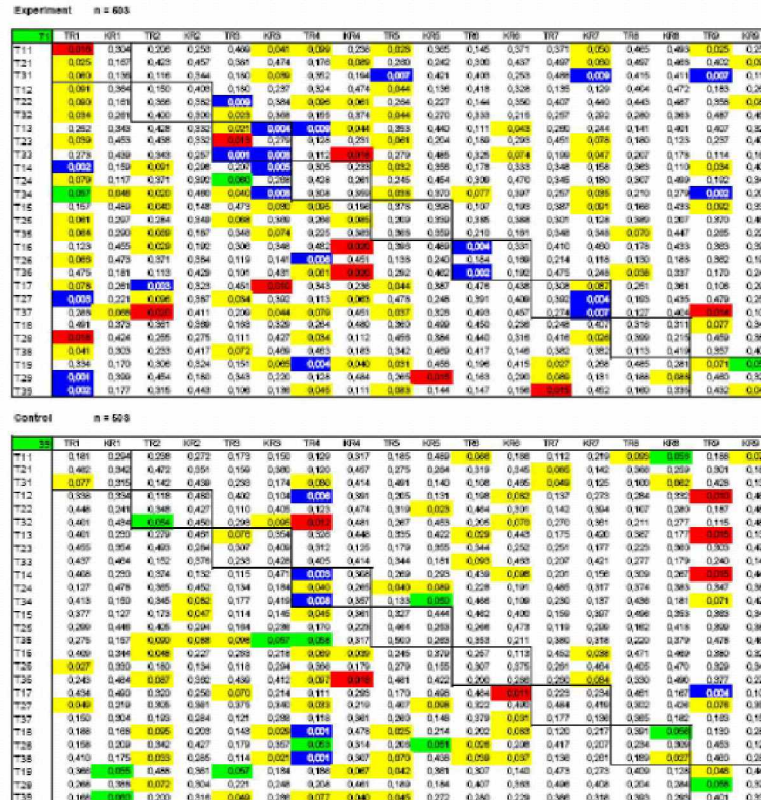


Figure 6: Correlationsmatrices CE und CC of the Viadrina-study (27*18 Matrix)

But there are correlative correlations: Each cell in the CE matrix is the result of a correlation measurement in which the correlation between the values of the psychological variables and those of the physical variables is determined for each of the 503 test subjects. Normally, at a certain level of significance, it is possible to calculate quite accurately how many cells in the matrix show significant correlations at random. But because we do not know the distribution function, a control matrix CC is generated. For this purpose, the experiment is run without test persons to generate the physical control data, but the same psychological data of the test persons is used for the control matrix. The two matrices CE and CC can now be compared with each other, and here one can see with the naked eye in Figure 6, even without statistics, that the number of correlations is significantly greater than in the control measurement CC. The blue cells are the more significant ones than the red ones. But in principle this does not matter now. It can be seen that this is a relatively simple method.

Now one can compare this with an EPR situation in physics (Figure 7). In physics you can specify the correlation function. There is a difference between the classical correlation function and the quantum mechanical one. It is mirrored in the figure because only the magnitude of the correlation is of interest. In the case of CMM, this can also be represented in this way. For all p-values (significance of the respective matrix cell) from 0 to 1, the number of cells for the experimental matrix CE and the control matrix CC is plotted. The lower curve is that of the control matrix and the upper curve is the experimental values. This is of course analogous to physics. It can be seen that in both cases the entanglement correlation is stronger than the classical correlation. One thinks that entanglement correlations are something fleeting, nothing

clear, but that is not true. Entanglement correlations are stronger than the classical ones because they have the "support" of the classical correlations in the OCS.

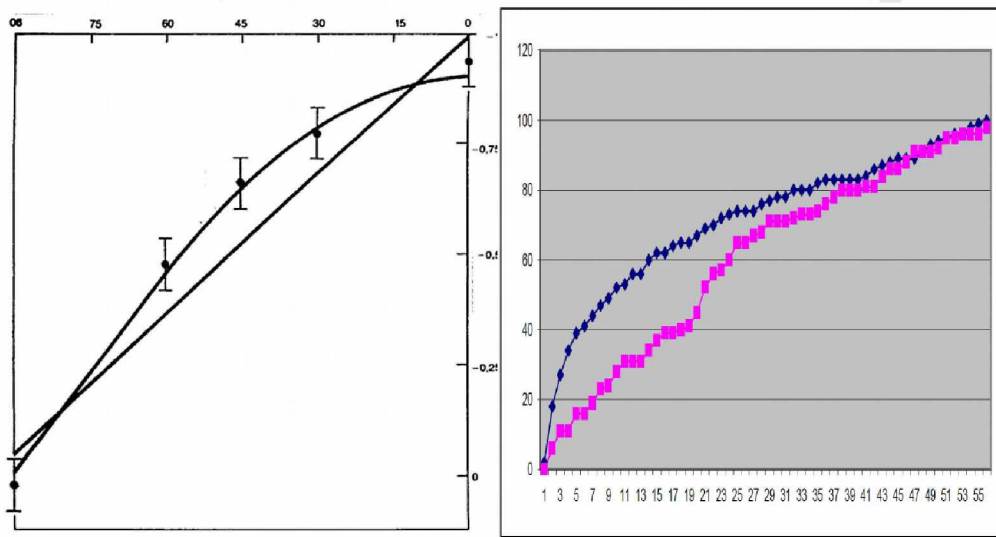


Figure 7: EPR-correlation-function in physics and CMM

In the described experiment, the instruction and the experimental setup serve to generate organisational closure. There is the test subject and the random generator; both have to be brought together somehow, so that they get organizationally closed. This is the art of the experimenter, one can do it better than the other and therefore there is certainly an experimenter effect.

Nevertheless, it is not the case that everyone who evaluates the data afterwards also has an influence on it. It only depends on the organizational closure of the respective system. This is important because in parapsychology there has long been speculation - and this has led to grotesque situations - whether the readers of a publication could still change the results of a parapsychological experiment. The entanglement correlations viewed from the outside are therefore no "experimenter effects"!

The first experiment was carried out in Freiburg. The test participants were really very motivated, many came specially and wanted to prove their psi abilities. The Viadrina study was an identical replication with the same variables. Nevertheless, there is an important difference between the two: The individual correlations lie at different points in the matrix.

This is a consequence of the NT axiom. If the correlations would occur at exactly the same position, then this would be a signal, and then the NT axiom would be violated. It ensures that you never get the same result, but only the difference between experimental and control matrix. But now the exciting question arises: I can also say that the actual effect is the difference and that is my "signal". But this is also subject to the NT-Axiom. Thus, the CMM effect would also become weaker and weaker during replication.

So what is the possibility for nature to maintain its entanglement correlation in CMM? You can easily think about that: Then the effect must appear in the control! This

means that an effect that I think I have under control suddenly appears in the control. This has also been established in homeopathic examinations. This means that the control is part of the organisationally closed system, so I cannot simply forget it in my mind. The entanglement "uses" every opportunity to escape the NT axiom.

So we have a pretty complicated context here that changes every time I do a new experiment.

Study	Subjects	Psych	Phys	Ncor	Nexp	Ncont	Nerw	p(limit)	Dmax	pCE-CC	pCP
Lucadou 1986 Reanalyse	299	63	32	2016	1413	1322	1371	0,60	91	0,0359	0,0000
Lucadou 1991 Reanalyse	307	43	56	2408	2019	1937	1999	0,83	82	0,6628	0,0000
Lucadou 2006	386	27	18	486	161	105	117	0,24	56	0,0268	0,2896
Walach et al 2014	503	45	45	2025	350	249	203	0,13	101	0,1003	0,0000
Ana Flores 2016	213	45	45	2025	609	340	405	0,20	269	0,0000	0,0733
Ana Flores 2017	200	45	45	2025	478	254	324	0,16	224	0,0016	0,0071
Ana Flores 2018	200	45	45	2025	569	371	324	0,16	198	0,0002	0,0981
Karolina Kirmse 2017	64	45	45	2025	907	805	830	0,41	102	0,1736	0,0013
Walach 2018 Clas RNG	22	45	45	2025	1526	1444	1458	0,72	82	0,2776	0,0747
Walach 2018 True RNG	15	45	45	2025	954	891	952	0,47	63	0,5319	0,3756
										1,1E-08	1,3E-12

Table 1: All CMM-Experiments until now, pCE-CC is the α -error, resulting from the permutation-method; pCP is the α -error of the CP-Method. N = number of correlations, Dmax = $D(p(\text{limit})) = N_{\text{exp}} - N_{\text{cont}}$, Nerw = expectation value for $p(\text{limit})$

Table 1 shows all experiments carried out directly according to this pattern. Dmax shows the maximum difference of the significant correlation between control and experiment. However, Dmax is not suitable as a test value to estimate the probability of the α -error because the psychological variables are not independent of each other. Hartmut Grote has proposed to use a Monte Carlo method. For this purpose, the physical variables are permuted and a distribution of the correlation frequency is determined. The α -error then results from the position of the experimental data (number of correlations) in this distribution. If one considers this correctly, however, one can show that this evaluation method adds a part of the correlations to the null hypothesis and is therefore very "unfavourable" or very conservative. Entanglement correlations do not only have the property to exist between psychological and physical variables, but they also have an effect on time. This means that the next test run also depends on the previous one and thus the permuted distribution widens. Unfortunately, we do not have another method yet. If one summarizes all ten studies, then at least a significance of $\alpha = 10$ to the power of -8 results; this is actually not so bad. For the evaluation of pCE-CC the permutation program of Bierman & Joly 2017 was used. Normally one would accept such a thing in psychology as proof of a connection.

A method (CP) for measuring the decoherence of entanglement

Nevertheless, there is now this constant annoyance with the psychological variables, because these are simply crooked and uneven. The keystrokes are not normally distributed variables. That's why I thought about using only the physical variables, not the psychological ones, for the CMM. The result is a single matrix CP consisting of

the physical variables of the experimental matrix CE and the physical variables of the control matrix CC. (I have announced this improvement before, so that it cannot be said that this is p-hacking) You have the test person who shows in an experiment what he or she can do, and there is a control for it. Now I throw out the psychological variables and have the possibility to test only the physical variables against the other physical variables from the control. This is an interesting question that has a long tradition in parapsychology, and it used to be called "the psychometric object". According to the GQT, this would be described as a kind of "macroscopic quantum excitation". Such "excited states", however, are subject to "decoherence", i.e. they decay after a certain time. We can check this now. We simply make several control measurements one after the other and check how long the correlations are still there. The values for CP in Table 1 are indeed much more significant than those for CMM. The Bierman & Joly 2017 program was also used to evaluate pCP. However, it can also be seen that for some it is not significant, and clearly not significant. So I took a look at the data and something interesting came out: In the first experiments here, I always did the control measurements intermittently. After each run there was a control run. In the other experiments, the control runs were all done at the end of the test sessions. The result was that exactly those that were intermittently performed showed the strongest effect. This is exactly what I expected. Because if I do a check right after each run and there's actually something like an excitation of the system that's slowly disappearing because of decoherence, that's where I'm going to expect the strongest effect. I think it's amazing. Whereas in the study, where the control runs were only carried out at the end of the whole study, there is no significant result at all.

The sheep are hiding on the riverbank

Hartmut Grote has pointed out that with this method it is extremely important to exclude causal artefacts from the random generator used. Such causal correlations can arise, for example, when different random generators are used in the study. However, these causal artefacts are stable cell populations in CP, which can be "picked out" if several control measurements are available. It is also possible to perform an extreme value analysis and omit all those that show a large effect in the individual test subjects until no effect can be seen in the end. Causal correlations remain stable almost to the end, while entanglement correlations fluctuate constantly. This can be used to test whether the correlations in CP are causal or entanglement correlations.

Hartmut Grote therefore recommended not using the "voltage variables" of the Viadrina study for CP, as such disturbances could occur as can be seen in Figure 8. I took this suggestion into account by subtracting the mean value of the voltage for the whole run from the voltage variables for each trial. As expected, the result is a reduction in significance, but it has not completely disappeared.

What does that mean? You could say it's possibly an artefact. This has to do with the fact that voltage values in random generators do not have a good image because they are difficult to keep stable. But even if you take that into account, the effect is still there. It doesn't disappear, it just gets weaker.

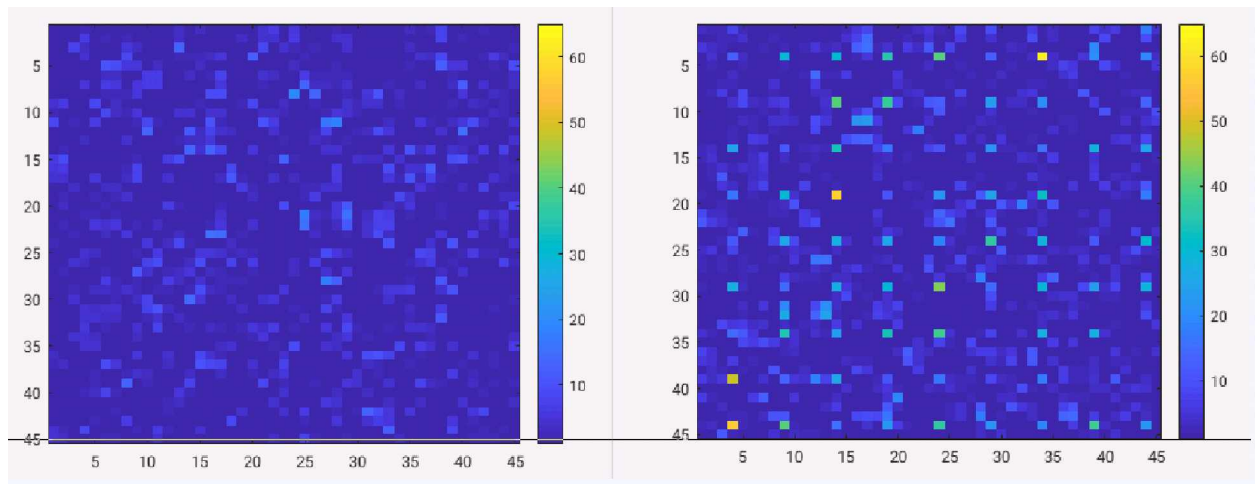


Figure 8: The CP-Matrix without and with causal artefacts (Hartmut.Grote: 16.01.2019)

To directly subtract the voltage is a rather massive intervention (I have faked my data with it to some extent.) This means for the OCS that the potentiality of a causal relationship has been reduced in a rabiant way. According to the NT axiom, the entanglement correlation must compromise between its establishment and its prevention. If the causal effects were cut out completely, they would stand "naked". And where are they going? After the GQT, one would expect them always to go where it is no longer possible to distinguish between causal and non causal effects. The sheep hide, so to speak, on the banks of the river. The rivers would then be the causal processes, and the sheep the Psi processes. And they prefer to hide where causal processes could also exist.

However, this means that we can investigate the interaction or dynamics that exist between causal and entanglement processes. In other words, nature always hides itself optimally - because of the NT axiom - where the NT axiom can no longer be easily damaged. In this case, the quality of the documentation is reduced; I cannot completely rule out the possibility that a causal artefact is present.

Are there any independent experiments in parapsychology at all?

On the basis of identical highly significant replication studies one can study the effect of the NT axiom very beautifully. For this purpose, the maximum deviation D_{max} between the matrices to be compared is considered (Table 2). The first three lines show the D_{max} values for the three studies according to the CMM (blue curve, Table 2): The psychological variables are compared with the physical variables of the experiment and the control runs. The D_{max} values show a slight decline effect. If one now permuts the respective physical control variables of the three studies, one obtains the next six rows in table 2. One can see that there is still a clear, but somewhat smaller deviation D_{max} . This indicates that the controls (violet curve, Table 2), as expected, are very close to the expected value of the number of correlations (yellow curve, Table 2) in the matrix. It can also be said that the control matrix indeed represents the part of the system that is largely free of entanglement. The significantly lower value of D_{max} in Study 2 can be made understandable - without going into the details - by the effect of the NT axiom. Here the control matrix shows the strongest negative deviation from the expected value (see above).

Study1	Study1	Study1	269
Study2	Study2	Study2	224
Study3	Study3	Study3	198
Study1	Study1	Study2	195
Study1	Study1	Study3	208
Study2	Study2	Study1	138
Study2	Study2	Study3	139
Study3	Study3	Study1	207
Study3	Study3	Study2	265
Study1	Study2	Study3	68
Study3	Study2	Study1	47
Study2	Study3	Study1	31

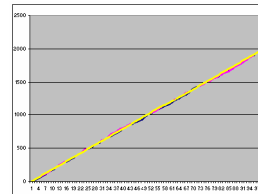
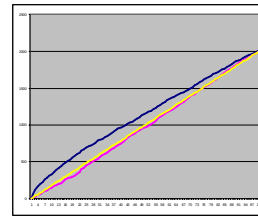


Table 2: Comparison of Dmax of three identical replication studies (blue = psychological data; red = physical data; black = control data)

The upper figure shows the typical course of the cumulative distribution $N(p)$ for the experimental data (blue), the expected value (yellow) and the control data (violet). The figure below shows the corresponding typical distributions for the last three rows of the table (CP matrices).

In the last three rows of Table 2, only the experimental physical variables of the three studies were permuted among themselves in a CP matrix. In all three cases, Dmax no longer shows a significant deviation from the expected value. If there were causal RNG artefacts in these three studies, they would have to be clearly visible here. Rather, the results show that the clear effects in the first three lines - especially with the voltage variables (see above) - were not caused by RNG artefacts. In addition, the result supports the decoherence assumption from above: Each study represents an independent "macroscopic quantum excitation" of the OCS. However, the effect of the NT-axiom is by no means completely eliminated, as we will see from the voltage variables. These are "highly charged" (see Figure 9). I now compare them directly by comparing the corresponding matrix cells individually. The NT axiom says that they must not be in the same place. If you look at this now, even in cases where they are very significant (blue cells) there is no correlation at this point in the following study (Figure 9). In the next replication study, however, the space is "free" again because the NT axiom can no longer be violated. This can be clearly seen in Figure 9 in the lower two matrices, although there is still a "certain shyness" that "the sheep hide here again". This, of course, proves that there can be no causal correlations. Because causal correlation would of course remain exactly in the same place.

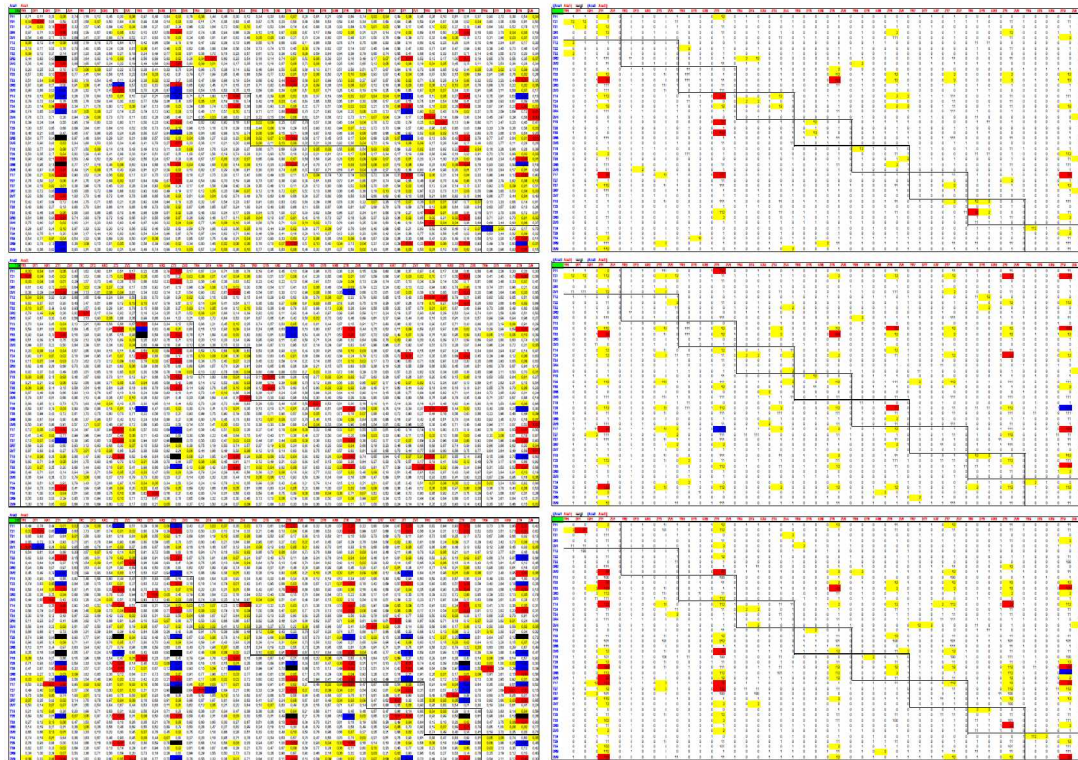


Figure 9: The correlation matrices CE of three consecutive replication studies (left matrices) and their comparison CP (right matrices). The first one is the difference between Study 1 and Study 2; the middle one compares Study 2 and 3, the lower one Study 1 and 3. Yellow corresponds to $p = 0.1$; red: $p = 0.01$; blue $p = 0.001$ and black: $p = 0.0001$.

In summary, based on the GQT, the MPI and the experimental findings, it can be said that in parapsychology only limited independent experiments are possible. Although they are subject to decoherence and the NT axiom, this also opens up a completely new perspective not only in parapsychology, but in all areas of the human sciences.

The Role of CMM and CP in the Human Sciences

The CMM and CP methods are of course by no means limited to parapsychology. In general, the aim is to simultaneously record and statistically evaluate causal and entanglement correlations of arbitrary systems. If these methods were used, for example, in the medical field, a number of psychological, physiological or medical variables could be collected before, during and after treatment and the data evaluated in a correlation matrix. For example, the homeopathy study in Figure 10 clearly shows the effect of the NT axiom. One can then find out which correlations are causal and which proportions are entanglement correlations. In medicine, these are usually referred to as "placebo" or "nocebo" effects. However, it is possible to determine very precisely on which factors these depend and why they often show very specific medical effects.

Randomised controlled trial of homoeopathy versus placebo in perennial allergic rhinitis with overview of four trial series

Morag A Taylor, David Reilly, Robert H Llewellyn-Jones, Charles McSharry, Tom C Aitchison

BMJ 321, 471-476.

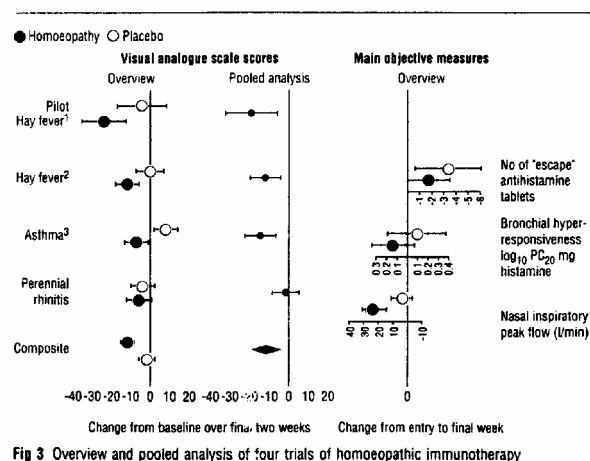


Figure 10: Four Homoeopathy-Studies in comparison

Figure 11 shows how causal and entanglement correlations can be separated. The figure is a simulation in which a causal process (the two left matrices) has been added to the data containing entanglement correlations. This requires two comparable matrices, as shown in Table 1. These can be obtained by using the data of two identical replications or by dividing the data into two equal data sets, so that the first line of data is used for the first data set and the following line - and so on - is used for the second data set. It is then necessary to define a suitable filter function with which the individual matrix cells of both matrices M1, M2 can be checked for correspondence. In figure 11 the following filter (in EXCEL) was used:

$$F(i,j) = IF(M1cell(i,j)<p;1;0)+IF(M2cell(i,j)<p;1;0)$$

p = significance of cell (i,j)

This filter generates the numbers from $F(i,j) = 0$ to 2 depending on the degree of correspondence. All cells where $F(i,j) = 2$ are causal correlations. All cells where $F(i,j) = 1$ are entanglement correlations. It is recommended to take a value for p where the difference between the two distribution functions (Bose-Einstein and Maxwell-Boltzmann; see below) is as large as possible. In Figure 11, the values of $p = 0.001$ (blue), 0.01 (red) and 0.1 (yellow) were used. The upper right matrix represents the causal correlations and the lower right matrix the entanglement correlations.

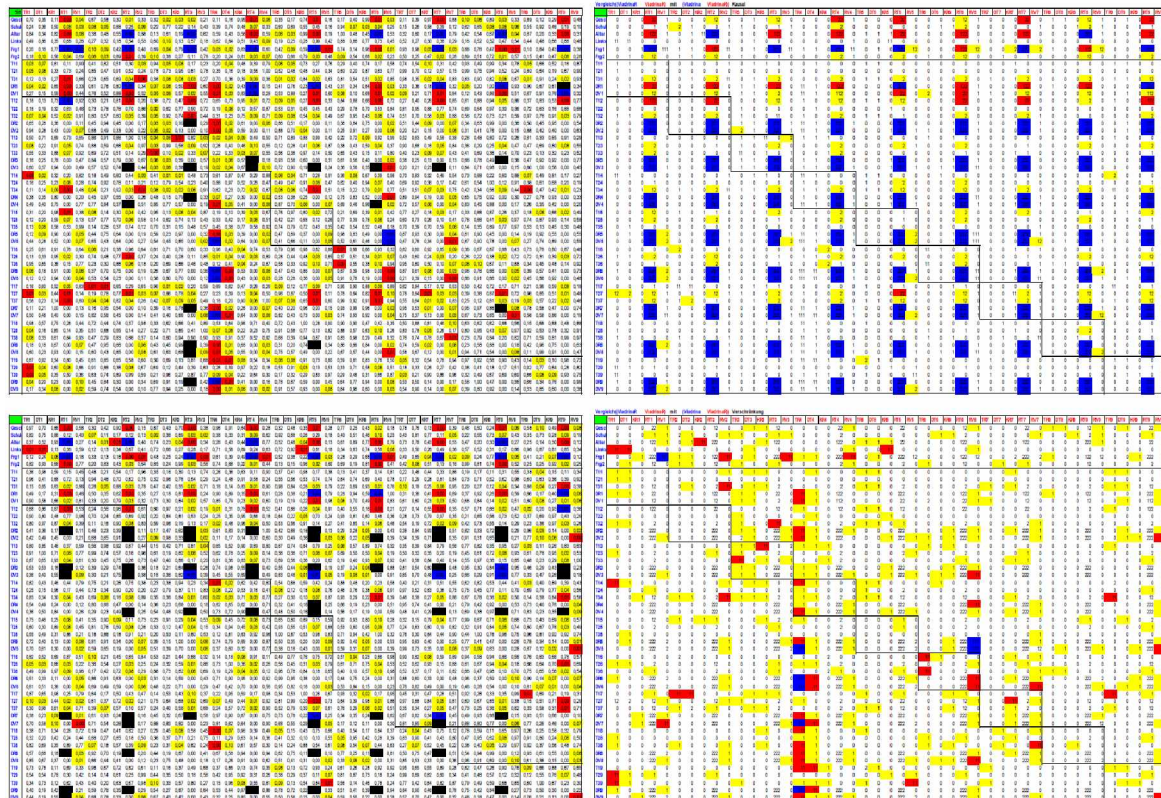


Figure 11: Separation of causal and entanglement correlations. The two left matrices show two simulated CMM replications. The upper right matrix shows the causal correlations, the lower right matrix the entanglement correlations.

In nature, causal and entanglement correlations in OCS are generally always common. They generate and stabilize each other. They are comparable to sponge and water. A dry sponge is something rigid. You can't do much with it. The water alone does not do much good, it flows away, is unstable. But if you bring both together, then you can clean properly with it. Macroscopic entanglement correlations are therefore ecologically stable, i.e. they do not simply go away, but are only limited by the NT-axiom. And they are formed by causal processes, which in turn stabilize the entanglement. In parapsychology, however, the experimental setting tries to omit the "sponge" and only catch the "water".

In our culture only the "causa efficiens" are taken into account. Aristotle has known three other forms of "causality", which - with certain restrictions - could be described as entanglement contexts. These are space-time independent, they also have a potential effect; they cannot be used to transmit signals.

For example, when a patient takes a homeopathic remedy, and there are many who say, "I don't believe it, but it works anyway" - they have the impression that there is a causal connection. And there's no reason why they shouldn't believe it. Because that strengthens the entanglement effect. This means that the wrong model is an advantage in this case. I call this a "pseudo signal" (see above). Pseudo-signals are generally stronger than causal signals, and they have the property of constantly reorganizing themselves. They are extremely flexible, fluid, and responsive to every change in the overall system, but we cannot deal with them as we do with causal processes.

Temporal clustering and synchronicity

There is a lot of people who say there's no coincidence. Often these are people who have experienced synchronistic events (in the sense of W. Pauli and C.G. Jung) or a "Kairos moment". A characteristic of such events is that they seem to occur "cumulatively" on the time scale. This is often dismissed as "selective" perception because we have the intuitive notion that random events must be evenly distributed over the time axis.

However, it could also be that there is entanglement behind it. As we have explained above, spatial entanglement in a system always leads to a temporal entanglement because of the NT axiom. This is expressed by the fact that the events on the time axis are no longer completely independent of each other, but form "clusters", i.e. accumulations. In (quantum) physics this is expressed by two different distribution functions. The Maxwell-Boltzmann distribution is "responsible" for normal coincidences, while the Bose-Einstein distribution describes processes that are "not distinguishable" or "not markable". The latter is especially true for entangled systems. The occupation of matrix cells in CMM cannot be marked in entanglement correlations because they can "jump" to another cell at any time, whereas causal correlations are "fixed".

The same can be said of the pragmatic information: it cannot be marked because the meaning of a thing does not depend on its name. For example, the words "blau", "bleu" or "blue" can be marked, but their meaning is not, it is the same in all cases.

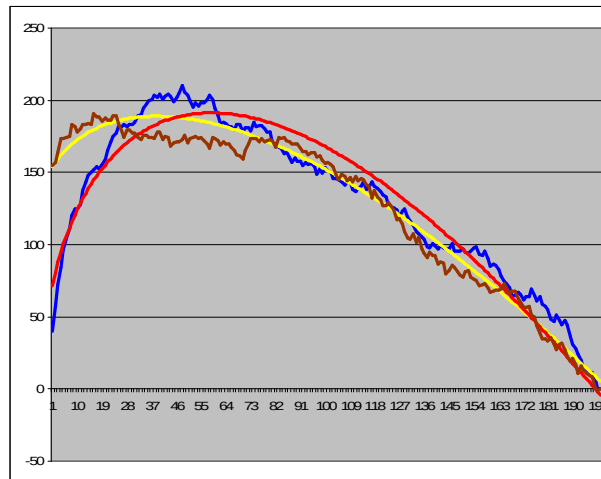


Figure 12: Adjusted cumulative Maxwell-Boltzmann (yellow) and Bose-Einstein distribution (red). The blue curve is the cumulative distribution of CMM Study 1 (Table 2) and the brown curve is the cumulative distribution of simulated causal correlations (lower left matrix from Figure 11). The linear expected value of the cumulative distribution (yellow curve in Table 2) for $p = 0, \dots, 200$ is subtracted for all curves.

From all this it follows that the MPI and the GQT predict that the distribution function for CMM cell contents (p -values) is distributed Bose-Einstein-like ($A+M/(EXP(p^E)-1)$) precisely when entanglement correlations are present, but otherwise Maxwell-Boltzmann-like ($A+M/(EXP(p^E))$). It is therefore to be expected that synchronistic events (Lucadou et. al. 2007) and entanglement correlations follow a Bose-Einstein-

like distribution, while causal correlations are markable and thus Maxwell-Boltzmann-like.

This was demonstrated by a single highly significant CMM study and the above causal simulation data. Figure 12 shows the corresponding distribution functions of the matrix elements (p-values). The Bose-Einstein distribution function was adapted with the values $M = 13.53$; $E = 0.14$; and $A = 60.00$; the Maxwell-Boltzmann distribution function with $M = 22.60$; $E = 0.14$; and $A = 150.00$. However, it is not possible to find parameters M , E , A that agree with the empirical data and allow an exchange of the two distribution functions. This might allow for a statistical distribution test for CMM correlations.

Of course, further investigations are necessary here. Should it turn out, however, that these results can be replicated, this would be a building block for the statement of the GQT that there are actually macroscopic entanglement correlations in psychophysical systems.

Literature

- Atmanspacher, H., Filk, Th., Römer, H. Quantum Zeno Features of Bistable Perception. *Biological Cybernetics* 90. 2004. pp. 33-40.
<http://arxiv.org/abs/physics/0302005>.
- Atmanspacher, H., Römer, H. & Walach, H. Weak quantum theory: Complementarity and entanglement in physics and beyond. *Foundations of Physics*. 2002; 32: pp. 379–406. <http://arxiv.org/abs/quant-ph/0104109>.
- Bierman, D., Jolij, J. (2017): Global Connectivity analysis program, available at: Heymansgroup.nl/publicsoftware
- Bösch, H., Steinkamp, F., Boller, E. (2006), "Examining psychokinesis: The interaction of human intention with random number generators: A meta analysis", *Psychological Bulletin*, 132 (4), pp. 497–523.
- Brunswick, E. (1956). *Perception and the representative design of psychological experiments*. Berkeley: University of California Press.
- Flores, A. , Watt, C., Tierney, I. Replication and Extension of a New Paradigm for Psychokinesis: Mind-Matter Interaction through Non-Local Entangled Correlations?. 41st International SPR Conference; 2017 September 1-3; De Vere Horsley Estate, East Horsley, KT24 6DT.
- Flores, A., Tierney, I. & Watt, C. Where mind connects with matter: Replicating the correlation matrix method. Paper submitted for presentation at the 61st Annual Convention of the Parapsychological Association August 2-5, 2018 Institute of Noetic Sciences, Petaluma, California.
- Kirmse, K. A. (2018): *Matrix Reloaded: Replikation des Matrix-Experiments mit Variation der psychologischen Variablen* (unpublished MA-thesis). Technische Universität Chemnitz; 2018.
- Lucadou, W.v. (1986): *Experimentelle Untersuchungen zur Beeinflussbarkeit von stochastischen quantenphysikalischen Systemen durch den Beobachter*. Frankfurt: H.-A. Herchen Verlag; 1986.
- Lucadou, W.v. Keine Spur von psi - Zusammenfassende Darstellung eines umfangreichen Psychokineseexperiments. *Zeitschrift für Parapsychologie und Grenzgebiete der Psychologie*. 1986; 28: pp. 169-197.

- Lucadou, W.v. (1987). A multivariate PK experiment, Part III. Is PK a real force? The results and their interpretation. *European Journal of Parapsychology*, 6, (4), 369-428.
- Lucadou, W.v. (1991). Locating psi-bursts - correlations between psychological characteristics of observers and observed quantum physical fluctuations. In: Delanoy, D.L. (ed.) *The Parapsychological Association 34th Annual Convention, Proceedings of Presented Papers*. Heidelberg: Parapsychological Association; 1991. pp. 265-281.
- Lucadou, W.v. (2002): Die Magie der Pseudomaschine. In: Belschner, W., Galuska, J., Walach, H., Zundel, E. (eds.). *Transpersonale Forschung im Kontext. Transpersonale Studien 5, Bibliotheks- und Informationssystem der Universität Oldenburg*. Oldenburg: 2002; pp. 77-100. also in: *Zeitschrift für Parapsychologie und Grenzgebiete der Psychologie*. 2009; 44/45/46: pp. 3-19. (English Version: *The Magic of Pseudo-Machines - Pk and other Oddities*. Invited paper presented at the Parapsychological Association 45rd Annual Convention. 2002 August 4-8; , Paris, France.
- Lucadou, W.v. (2006): "Self-organization of temporal structures - a possible solution for the intervention problem", *Proceedings of the 87th Annual Meeting of the AAAS Pacific Division, University of San Diego June 18-22, 2006, Symposium: Frontiers of Time: Reverse Causation – Experiment and Theory*. Daniel P. Sheehan (ed.): *Frontiers of Time. Retrocausation - Experiment and Theory*. AIP Conference Proceedings, Volume 863, AIP, Melville, New York, 2006, 293-315.
- Lucadou, W.v. (2015a): The Model of Pragmatic Information (MPI). In: May, E.C., Marwaha, S. (eds.). *Extrasensory Perception: Support, Skepticism, and Science. Theories and the Future of the Field*. Santa Barbara, USA: Praeger publications; 2015; 2., pp. 221-242.
- Lucadou, W.v. (2015b): The Correlation-Matrix Method (CMM) - a New Light upon the Repeatability Problem of Parapsychology. Paper for the 58th Annual Convention of the Parapsychological Association and 39th SPR International Annual Conference. 2015 July 16-19. University of Greenwich, Great-Britain. *Journal of Parapsychology*. 2015; 79 (2): pp. 145-146.
- Lucadou, W.v. (2018): The Model of Pragmatic Information (MPI) and Generalized Quantum Theory (GQT). *Mindfield* 10.3 2018, p. 109 - 115.
- Lucadou, W.v. (2019): Homoeopathy and the Action of Meaning - A Theoretical Approach, Accepted for publication. *Journal of Scientific Exploration*.
- Lucadou, W.v., Römer, H, Walach, H. (2007): Synchronistic Phenomena as Entanglement Correlations in Generalized Quantum Theory, *Journal of Consciousness Studies* 14, 4(2007), pp. 50-74.
- Primas, H. (1990): Mathematical and philosophical questions in the theory of open and macroscopic quantum systems. In: Miller, A.I. (ed.). *Sixty-two years of uncertainty*. New York: Plenum; 1990. pp. 233-257.
- Primas, H. (1992): Time-asymmetric phenomena in biology complementary exophysical descriptions arising from deterministic quantum endophysics. *Open Systems & Information Dynamics*. 1992; 1: pp. 3-34.
- Renninger, M. (1960): Messungen ohne Störungen des Messobjekts, in: *Zeitschrift für Physik* 158, 417-421.
- Schmidt, H. (1971): Mental Influence on Random Events. *New Scientist and Science Journal* 1971, 757-768.

- Taylor, M. A., Reilly, D., Llewellyn-Jones, R. H., McSharry, C., Atchison, T. C.
Randomized controlled trial of homeopathy versus placebo in perennial allergic rhinitis with overview of four trial series. BMJ 2000; 321: pp. 471-476.
- Timm, U. (2007): Reanalyse einer Metaanalyse von Rng Pk Experimenten. WGFP Workshop; 2007 October 10; Offenburg, Germany.
- Varela, F. J. (1981): Autonomy and autopoiesis. In: Roth, G. , Schwengler, H. (Eds). Self-organizing systems. Frankfurt/New York: Campus; 1981.pp. 14-23.
- Walach, H., Horan, M.(2014): Capturing PSI - Replicating von Lucadou's Correlation Matrix Experiment using a REG Mikro-PK. Presentation at the BIAL-Conference. 2014 March. Porto, Portugal.
- Walach, H., Horan, M., Hinterberger, T., Lucadou, W.v. (2019): Evidence for a Generalised Type of Nonlocal Correlations between Systems using Human Intention and Random Event Generators. Submitted for publication in "Psychology of Consciousness: Theory, Research, and Practice" in 16th of January 2019. 2016.
- Weizsäcker, E.v. (1974): "Erstmaligkeit und Bestätigung als Komponenten der pragmatischen Information", in: Weizsäcker, E. v. (Hrg.): Offene Systeme I. Stuttgart: Klett, pp. 83 113.
- Zuse, K. (1969): Der rechnende Raum 1969 by Springer Fachmedien Wiesbaden
Ursprünglich erschienen bei Friedr.Vieweg + Sohn GmbH, Braunschweig 1969

Author's address:

Dr.rer.nat. Dr.phil. Walter v. Lucadou
Parapsychologische Beratungsstelle
Hildastr. 64, D-79102 Freiburg i. Br.
Tel./Fax. +49-0761-77202
E-mail: info@parapsychologische-beratungsstelle.de